

Lessons Learned Report

Environment & Climate Analytics Dashboard Project

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Role: Project Manager & Owner (Temperature & AQI Analysis)
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1. Introduction

The **Environment & Climate Analytics Dashboard Project** focused on analyzing long-term environmental patterns across **Atlantic Canada (Nova Scotia, New Brunswick, Prince Edward Island, and Newfoundland & Labrador)**.

The project integrated multiple analytical domains, including:

- Temperature trends
- Air Quality Index (AQI)
- Precipitation
- Extreme weather events
- Greenhouse Gas (GHG) emissions

The objective was to provide a **holistic understanding of climate variability and environmental risk across the Atlantic region**, enabling both strategic and operational decision-making.

2. Project Scope & My Role

Project Scope

The dashboard was structured into **four major analytical domains** across Atlantic Canada:

Domain	Description
Temperature & AQI	Climate trends, anomalies, air quality behavior
Precipitation & Extreme Weather	Rainfall patterns, variability, extreme events
GHG Emissions	Provincial and sector-level emissions
Integrated Insights	Cross-domain correlations and risk patterns

My Role

I contributed in two critical roles:

1. Project Manager

As Project Manager, I was responsible for:

- Coordinating work across **multiple analytical domains**
- Managing dependencies between:
 - Temperature
 - AQI
 - Precipitation
 - GHG modules
- Ensuring timely delivery of integrated outputs
- Facilitating collaboration across team members
- Handling unexpected challenges (timeline delays, dependencies, changes)

2. Domain Owner – Temperature & AQI

I was the **primary owner of the Temperature and AQI analysis**, which included:

- Designing KPI metrics (temperature trends, AQI levels, unhealthy days)
- Performing correlation analysis between:
 - Temperature and AQI
 - Weather variables and pollution
- Identifying insights such as:
 - **0.31°C/decade warming trend**
 - **166 unhealthy AQI days despite “Good” average**
- Building interactive dashboards and narrative insights

👉 My module was a **core component of the overall dashboard**, contributing significantly to final insights and decision-making value.



3. Executive-Level Learnings



Cross-Domain Insight Integration

A major learning as Project Manager was:

“Insights are most valuable when multiple domains are integrated.”

Examples:

- Temperature + AQI → explains health risks
- Temperature + precipitation → shows variability
- GHG + temperature → explains long-term climate drivers

👉 Single-domain analysis is limited; integrated analysis provides **true business value**

5. Technical Learnings

Multi-Domain Data Integration

The project required combining datasets across:

- Climate observations (~997K records)
- AQI and pollutant data (multiple large files)
- GHG emissions
- Precipitation and extreme weather events

👉 This reinforced:

- Importance of **data consistency**
- Need for **cross-domain modeling**
- Complexity of integrating environmental datasets

6. Team, Leadership & Project Management Learnings (Critical Reflection)

6.1 Teamwork in a Multi-Domain Project

One of the most important lessons learned was the complexity of working in a **multi-domain team environment**, where each member owned a different analytical area (Temperature & AQI, GHG, Precipitation, Extreme Events).

Challenges included:

- Aligning outputs from different domains

- Ensuring consistent logic and interpretation across modules
- Managing dependencies between datasets

👉 Key Learning:

“A successful BI project requires strong alignment across all components — not just individual contributions.”

6.2 Conflict Resolution & Decision Alignment

The team frequently encountered differences in:

- Dashboard design
- Analytical approach
- Interpretation of insights

As Project Manager, it was important to:

- Facilitate discussion openly
- Focus on **project goals rather than personal preferences**
- Resolve conflicts by prioritizing **business value and clarity**

👉 Key Learning:

“Effective conflict resolution is not about winning an argument — it is about aligning towards the best outcome for the project.”

6.3 Leadership Under Pressure

The project presented real leadership challenges:

- Managing **tight timelines**
- Handling **last-minute changes** in requirements
- Dealing with **unexpected team member unavailability**

This required:

- Quick decision-making
- Redistributing responsibilities
- Maintaining team morale

👉 Key Learning:

“Leadership is demonstrated most clearly during uncertainty and pressure, not when everything goes as planned.”

6.4 Managing Shared Tools & Collaboration (PBIX Challenges)

Working with Power BI introduced practical challenges:

- Limited version control in **PBIX files**
- Risk of overwriting work
- Difficulty in collaborative editing

👉 This required:

- Pre-planning file sharing
- Clear ownership of sections
- Strict coordination during updates

👉 Key Learning:

“Technical tools can create collaboration constraints — planning workflows is as important as technical skills.”

6.5 Handling Sudden Changes & Adaptability

The project required adapting to:

- Evolving requirements
- Integration issues across domains
- Final-stage refinements before presentation

👉 Key Learning:

“Flexibility and adaptability are critical in real-world projects, where conditions change continuously.”

6.6 Time Management & Workload Planning

A major realization from this project:

- **Data cleaning and ETL consumed the majority of time (~60–70%)**
- Visualization and insights development required parallel effort

Challenges included:

- Underestimating effort for data preparation
- Managing parallel tasks across team members

👉 Key Learning:

“Effective time management requires prioritizing data preparation early and planning all phases realistically.”

6.7 Understanding the End User

Another important lesson was recognizing that:

- Not all users need technical information
- Insights must be tailored to different audiences

Audience	Focus
Executive	Strategy, ROI, risk
Business Users	Operational actions, usability

👉 Key Learning:

“Understanding the end user is critical — insights must be designed for how they will actually be used.”

6.8 From Data to Real-World Impact

The project emphasized that:

- Tools and data alone do not create value
- Insights must lead to **real-life impact and decisions**

Examples from the dashboard:

- AQI spikes → public health risk
- Heatwaves → operational disruption
- Climate variability → infrastructure strain

👉 Key Learning:

“The true value of BI lies in converting data into actionable decisions and real-world impact.”

6.9 Teamwork & Personal Growth

This project contributed significantly to professional development:

- Improved communication skills
- Enhanced collaboration abilities
- Developed leadership mindset
- Learned to manage diverse opinions

👉 Key Learning:

"Teamwork is not just collaboration — it is the ability to align diverse perspectives into one coherent outcome."

7. Analytical Learnings

Regional vs Local Insight

Key learning:

"Climate analysis must balance regional patterns with local variations."

- Atlantic Canada shows **overall warming trends**
- But provinces behave differently (e.g., PEI vs NS)

👉 Insight:

- Strategy should be **regional**
- Actions should be **localized**

REFLECTION

"This project demonstrated that successful Business Intelligence delivery requires more than technical skills.

It requires the ability to integrate data across domains, manage teams effectively, resolve conflicts, and translate insights into meaningful business impact.

As a Project Manager and domain owner, this experience strengthened my leadership, problem-solving, and strategic thinking skills while demonstrating how data can drive real-world decisions."